

A STUDY ON THE INFLUENCE OF SHAPE OF COARSE AGGREGATES ON STRENGTH OF DENSE BITUMINOUS MACADAM

¹Mr. Sumanth S, ²Mr.Suhas R, ³Dr.S.P.Mahendra

¹PG Student, ²Assistant Professor, ³Professor

^{1,2}Department of Highway Technology,

^{1,2}Dayananda Sagar College of Engineering,
Bengaluru, India

³Department of Civil Engineering,

³P.E.S. College of Engineering
Mandya, India

Abstract: In Hot Mix Asphalt (HMA), aggregates are consolidated with an asphalt binding medium to shape a compound material. By weight, aggregates constitute about 92 to 96 percent of HMA. They include the larger part of pavement volume. Therefore, it is important to know the knowledge of property of these aggregates in designing high quality pavements. Aggregates particle shape is one such property that is analyzed in this dissertation work. The shape of aggregate is categorized by measuring particles largest diameter, smallest diameter and Intermediate diameter using Vernier Calipers. With these diameter the elongation ratio and flatness ratio is calculated with which the aggregate shape is determined. Later the Aggregate gradation required for Dense Bituminous Mix is prepared as per MoRTH guidelines. For this dissertation work a proportion of 10%, 20%, 30% 40% and 50% of different shape of aggregates is prepared and mix strength is determined using Marshall Stability Test. Also, other marshall properties such as flow value, bulk density VFB and VMA is analyzed. For the mix with maximum strength and minimum strength is analyzed for its performance evaluations using Immersion Wheel tracking equipment. From these dissertation work it is observed that more the sphericity value of aggregates higher it resembles the cubical shape of aggregate. In DBM mix cube shape of aggregates with 20% has the higher strength than conventional mix, followed by strength of 50% cube shape aggregate replacement in conventional mix. Blade shape aggregate with 30% in the mix has the least strength. Coming to performance evaluation cube shape with 20% proportion has more rutting resistance potential than the conventional mix.

Keywords—Hot Mix Asphalt, Dense Bituminous Mix, MoRTH, Marshall Stability, Aggregate Shape, Sphericity, Immersion Wheel Tracking.

I. INTRODUCTION

Since around 95 percent of the aggregate volume of HMA blends comprises of aggregates, the execution of HMA blends is incredibly influenced and impacted by the properties of the total mix. The total properties that fundamentally impact the execution of HMA blends are gradation, shape (angularity) and texture (roughness). The shape and texture of coarse and fine aggregates control the blend's quality and rutting resistance in this way influencing execution and serviceability. Rough angular precise aggregates create higher quality pavements than smooth uncrushed aggregates. Aggregate shape can be portrayed as cubical, flat, elongated and round. The nearness of flaky aggregates is considered as undesirable in bituminous blends in light of their propensity to separate amid development and ensuing movement operations. The voids display in a compacted blend rely on upon the state of aggregates. Exceedingly flaky aggregates have more voids and diminish the workability. Henceforth it was felt that the examination on the impact of the flaky aggregates on bituminous blends is significant and fundamental.

II. SCOPE AND OBJECTIVE OF THE STUDY

To study the performance and behaviour of influence of shape of coarse aggregates on strength of dense bituminous mix and also to determine the different individual properties of materials (aggregates and bitumen).

- a) To categorised the aggregates in to different shapes.
- b) To determine the behaviour of DBM mixes with aggregate having different shapes and different proportions (10%, 20%, 30%, 40%, and 50%).
- c) To determine the particle index of the coarse aggregates.
- d) The following properties of dense bituminous mix were determined in this study by conducting Marshall Tests. Marshall Stability, flow value, air voids (Va), Voids in Mineral Aggregate (VMA), Voids Filled with Bitumen (VFB).
- e) To determine the rutting potential of DBM mix with different shape proportion.
- f)

III. MATERIALS USED IN THE STUDY

- I. AGGREGATES:** The aggregates selected for the Dense Bituminous Macadam are subjected to various aggregate tests as specified by IRC: SP: 79: 2008 and MoRTH section 500, confirming to the table 500-8. Good durable quality crushed aggregates of different sizes are obtained from quarry Q-con near Mandya.

Table 1 : Properties of Coarse Aggregates

SI NO	Aggregate tests	Standard results as per MORTH-500
1	Aggregate Impact Test	Max 27%
2	Specific Gravity	2.5-2.7
3	Flakiness index Elongation index Combined index	Max 30%
4	Abrasion test	Max 35%
5	Water Absorption	Max 2%
6	Crushing strength	Max 30%

- II. BINDER:** Bitumen (VG-30) was used as the binder in the mixture design of Dense Bituminous Macadam Mix throughout the dissertation work.

Table 2 : Properties of Bitumen

SI NO	Test Method	Standard results MoRTH-500 (IS:73)
1	Specific gravity test	1.02
2	Penetration test	Min 45 mm
3	Ductility test	Min 40 mm
4	Viscosity test	30 centi poise
5	Softening point test	45°C-55°C

- III. MINERAL FILLER:** The Stone dust has been used as the mineral filler in the Dense Bituminous mixture grading requirement of mineral filler are followed as given in MoRTH-500. The gradation requirements of mineral filler are given in Table 3.

Table 3 : Gradation details of Mineral Filler

IS SIEVE (mm)	Cumulative per cent passing by weight of total aggregates
0.6	100
0.3	95-100
0.75	85-100

IV. Quantification of aggregates

Coarse aggregates are classified in to different shapes using zingg's diagram classification of coarse aggregates a Aggregates passing 19mm and retained on 13.2 mm IS sieve is collected from sieve analysis. The collected samples is cleaned thoroughly and its largest diameter (d_l), smallest diameter (d_s) and intermediate diameter (d_i) is measured using Vernier Calipers. The elongation ratio and flatness ratio is calculated using the formula,

$$\text{Elongation ratio} = d_l/d_i$$

$$\text{Shape factor} = ds/\sqrt{d_i * d_l}$$

$$\text{Flatness ratio} = d_s/d_i$$

$$\text{Sphericity} = \sqrt[3]{ds * d_i/d_l^2}$$

With these ratios the aggregates are classified in to Blade shape, Disc shape, Cubical shape and Rod shape as shown in table 4

Table 4 :Aggregate shape classification using zingg's classification

Aggregate shape	Elongation ratio	Flatness ratio
Blade	$< 2/3$	$< 2/3$
Rod	$< 2/3$	$> 2/3$
Disc	$< 2/3$	$< 2/3$
Cubical	$> 2/3$	$> 2/3$

**Fig.1: Aggregates shape used in the study****V. RESULTS AND DISCUSSION**

- A. Aggregates** The aggregates were evaluated for various physical properties in accordance with the Indian Standard specifications. Table5 represents the test results of physical characteristics of aggregates used in the present dissertation work.
- B. Binder** Bitumen (VG-30) was used as the binder in the mixture design of Dense Bituminous Macadam Mix throughout the dissertation work. Table6 gives the experimental results.
- C. Mineral filler** The Stone dust has been used as the mineral filler in the Dense Bituminous mixture

Table 5: aggregate test results

SI NO	Aggregate tests	Results
1	Aggregate Impact Test	20.8%
2	Specific Gravity	2.6
3	Flakiness index	13.80%
	Elongation index	13.62%
	Combined index	27.42%
4	Abrasion test	24.81%
5	Water Absorption	0.2%
6	Crushing strength	28.46%

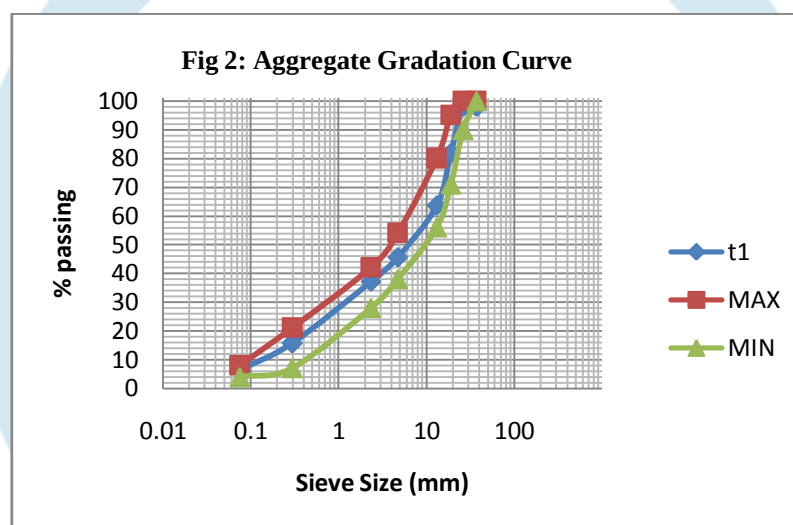
Table 6: Bitumen test results

SI NO	Test Method	Results
1	Specific gravity test	1.02
2	Penetration test	65mm
3	Ductility test	94mm
4	Viscosity test	30 centi poise
5	Softening point test	49°C

D. Aggregate Gradation

After the basic tests, the aggregates are further used in the determination of the proper blend to give a good mix consisting of different size of aggregates. The aggregate gradation is done by using Job Mix Formula Method to find the individual percentages of different sized aggregates to be used confirming to the upper and lower limits specified as per MoRTH table 500-10 grading-2. The gradation of aggregates for DBM-grade2 mix as shown in Table 7

TABLE 7 AGGREGATE GRADATION			
Material A: 19mm-12.5mm: 53%			
Material B: 12.5mm-2.36mm: 2%			
Material C: 2.36mm-0.075mm: 43%			
Material D: 0.075-PAN : 2%			
MoRTH specification			
Upper limit	Lower limit	Middle limit	Trail mix
100	100	100	98
100	90	95	98
95	71	83	82
80	56	68	64
54	38	46	46
42	28	35	37
21	7	14	16
8	4	6	6



E. **Particle Index Value Test** This test is used to determine the combined effect of particle shape and surface texture of aggregates. Aggregates of different shapes are tested individually in this experiment and result is tabulated Table 8 gives the experimental results.

Table 8 :particle index value

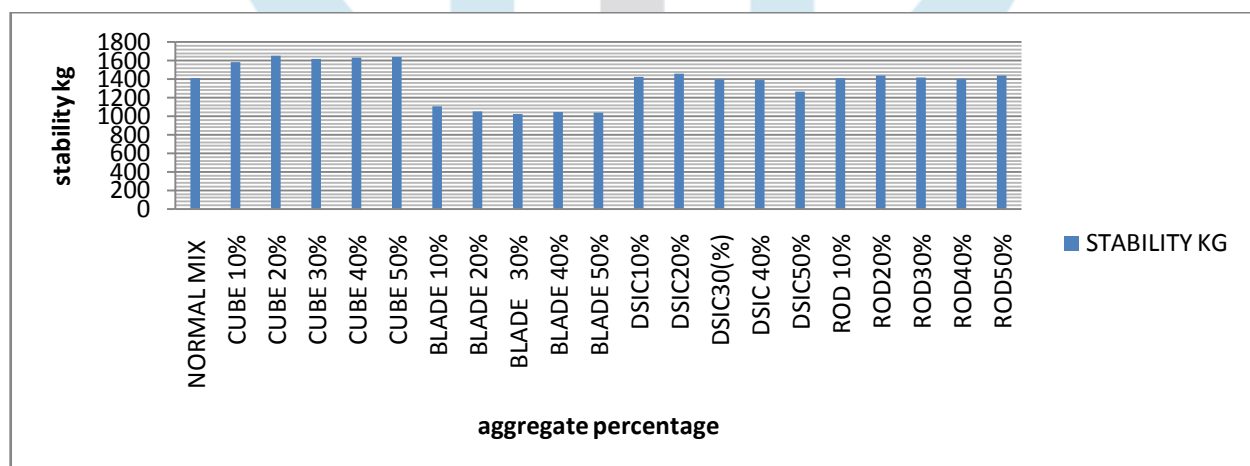
SL NO	SHAPE	M ₁₀	M ₅₀	V ₁₀	V ₅₀	P.I
1	Cube	4370	4515	46.86	45.09	15.3
2	Rod	4533	4744	44.87	42.31	13.5
3	Dsic	4676	4711	43.14	42.71	11.2
4	Blade	4855	4995	40.96	39.74	9.2

F. **Marshall Stability Test for different Shape and Proportion of Aggregates**

The Marshall specimens are prepared with different shape of aggregates (Blade, Cube, Rod and Disc) with different proportions (10%, 20%, 30%, 40%, 50%) in the mix. The blending of the mix remains same as of the conventional mix but the optimum bitumen content of 4.65% is maintained. For every shape and proportion of aggregates three samples is prepared and average stability value is considered. In this dissertation work it is observed that cube shape of aggregates with 20% replacement in the mix has highest strength and blade shape with 30% replacement has minimum strength. Following tables gives the average Marshall stability values for different shape and proportion of aggregates.

Table 9: Marshall Stability Results

MARSHAL TEST RESULTS	NORMAL MIX	CUBE 10%	CUBE 20%	CUBE 30%	CUBE 40%	CUBE 50%	BLADE 10%	BLADE 20%	BLADE 30%	BLADE 40%	BLADE 50%
STABILITY (Kg)	1405.35	1577.04	1650.62	1615.64	1627.52	1634.26	1103.88	1051.27	1019.37	1043.21	1033.64
FLOW (mm)	2.93	2.65	2.82	2.05	2.05	1.92	1.97	1.64	1.74	2.03	2.17
AIR VOIDS (%)	4.103	3.44	3.5	3.68	3.74	4.16	3.38	3.50	3.565	3.62	3.74
DENSITY (gm/cc)	2.49	2.48	2.41	2.42	2.41	2.42	2.403	2.42	2.41	2.42	2.43
VFB (%)	80.3	80.47	80.91	80.82	80.88	80.74	80.99	80.81	80.9	80.83	80.76
VMA (%)	12.65	12.71	12.63	12.62	12.61	12.57	12.64	12.65	12.63	12.63	12.63
MARSHAL TEST RESULTS	NORMAL MIX	DSIC 10%	DSIC 20%	DSIC 30%	DSIC 40%	DSIC 50%	ROD 10%	ROD 20%	ROD 30%	ROD 40%	ROD 50%
STABILITY (Kg)	1405.35	1420.32	1453.32	1393.27	1386.51	1262.26	1404.69	1432.20	1412.59	1403.19	1432.00
FLOW (mm)	2.93	2.13	2.03	1.98	1.93	1.90	2.19	2.17	2	1.82	1.53
AIR VOIDS (%)	4.103	3.44	3.62	3.80	3.92	4.1	4.16	4.16	4.10	4.16	3.92
DENSITY (gm/cc)	2.49	2.45	2.40	2.45	2.40	2.4	2.43	2.42	2.48	2.4	2.43
VFB (%)	80.3	80.93	80.65	80.9	80.58	80.84	80.64	80.72	80.34	80.85	80.68
VMA (%)	12.65	12.64	12.66	12.59	12.63	12.57	12.59	12.58	12.64	12.56	12.61

**Fig 3: Marshall stability variation**

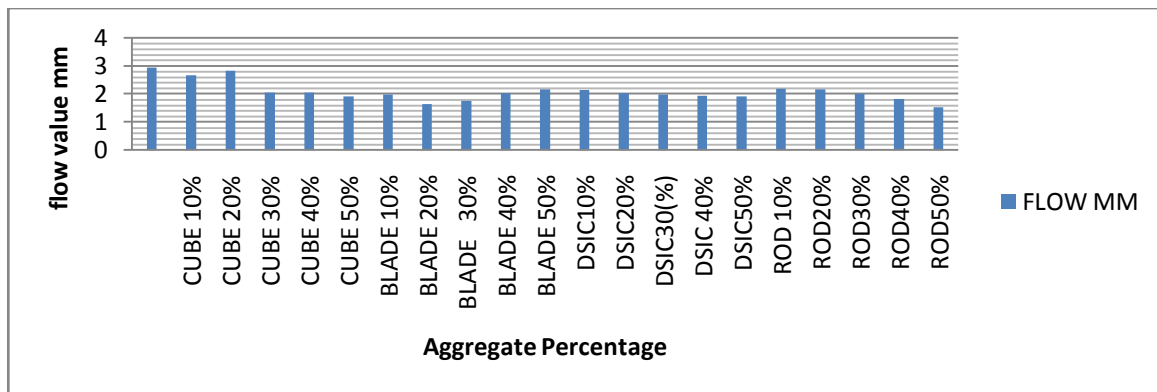


Fig4: Marshall Flow Value Variation

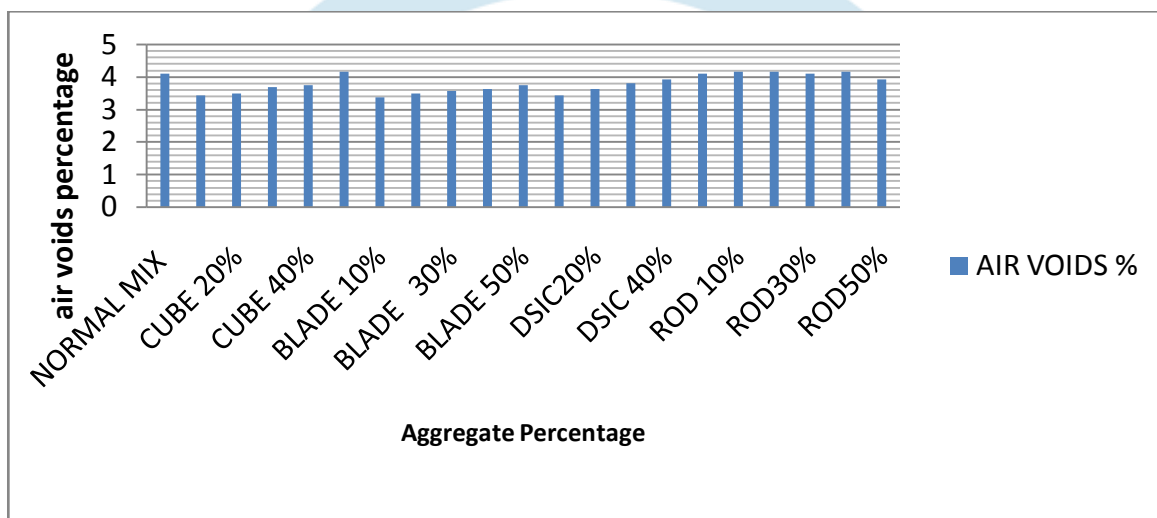


Fig5: Air voids percentage Variation

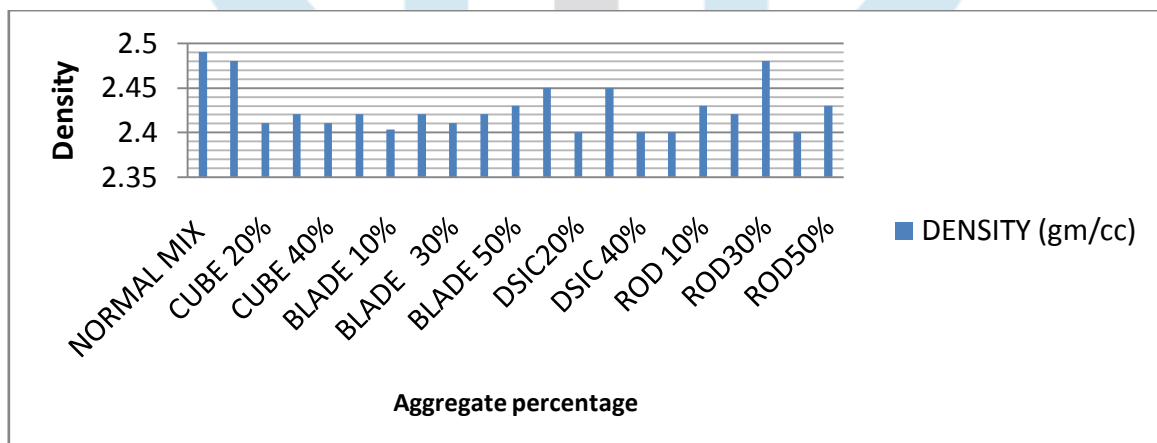


Fig6: Density Variation

G. Immersion wheel tracking The rutting test was done for both Conventional DBM mixture as well as with mixture having replacing with cube 20% and 50% since this mix has maximum strength as per Marshall stability test also blade with 30% replacement which has minimum strength. The test carried out for 60mm thickness with tyre pressure 7.2 Kg/cm² and the optimum binder content (OBC) being 4.65% and cube 20% mix with thickness 50mm.

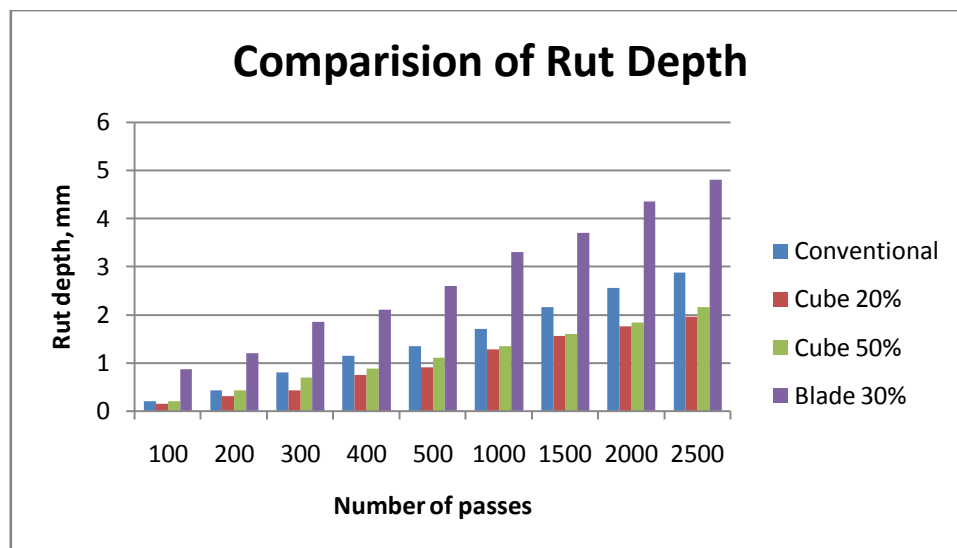


Fig6: Comparison of Rut Depth

VI. CONCLUSIONS

The followings are the conclusions based on test results:

- A. The aggregates and binder content test results are well in conference with specifications as per IRC: SP: 79: 2008.
- B. In particle index value test it is observed that higher the value of PI aggregate is cubical in shape, Rod shape aggregates having PI value slightly less than the cube shape, Dsic shape aggregates having PI value minimum and blade shape of aggregates being least in PI value.
- C. The stability of the Marshall mix with different aggregate shape and proportion shows the good results against the minimum Marshall strength of 9KN.
- D. In Marshall stability test Cube shape aggregates having 20% proportion in DBM mix possess higher stability of 1650.62 kg followed by cube with 50% proportion of strength 1634.26 kg.
- E. Presence of Rod shape of aggregates in the Marshall mix posses same strength as that of conventional mix of 1432 kg.
- F. Blade shape of aggregates in the mix has the least strength compared to other shape of aggregates strength in the mix. Blade shape with 30% proportion in the Marshall mix have less strength of 1019.37 kg.
- G. Mix prepared with 20% replacement cube, blade, rod and dsic aggregates shown higher stability values.
- H. In the performance evaluation i.e Immersion wheel tracking test conventional mix have the rut depth of 2.88mm by the end of 2500 passes whereas mix with maximum stability of cube-20 develop rut depth of 1.95mm there by the rut depth is reduced by 33% in cube-20 than that of conventional mix.
- I. Also the mix cube-50 has the rut depth of 2.15mm there by 25% of rut depth is reduced than that of conventional mix.
- J. Cube 20% mix with reduction in thickness by 10mm has the rutting potential to that of the conventional mix.
- K. Replacing the conventional mix with cube 20% in bituminous coarse reduces the cost of pavement with decrease in DBM layer thickness.

REFERENCE

- [1] **ASTM :D3398-00**, Standard test method for index of aggregate shape and texture
- [2] **Benson, J. R**The Grading of Aggregates for Bituminous Construction. ASTM Symposium on Mineral Aggregates, 1948, pp. 117-133.
- [3] **Boutillier, O.D.**(1967) A study of the relation between the particle index of the aggregate and the properties of bituminous aggregate mixtures. Proceedings of Association of Asphalt Paving Technologists, 36, 157-179.
- [4] **Brown, E.R., McRae, J.L, and Crawley, A.B.**(1989) Effect of aggregate on performance of bituminous concrete. ASTM STP 1016, Philadelphia, 34-63.
- [5] **Ganapatinaidu .p, s. adishesu** / International Journal of Engineering Research and Applications (IJERA) ISSN: 2248-9622 www.ijera.com Vol. 1, Issue 4, pp. 2013-2024 2024
- [6] **Huber, G.A., and Heiman, G.H.** (1987) Effect of asphalt concrete parameters on rutting performance: a field investigation. Proceedings of Association of Asphalt Paving Technologists, 56, 33-61.
- [7] **IRC:37-2012**, Guidelines for design of flexible pavements.

- [8] **IS 2386**, Method of test for aggregates.
- [9] **Kandhal, P.S., Khatri, M.A., and Motter, J.B.** (1992) Evaluation of particle shape and texture of mineral aggregates and their blends. *Journal of Association of Asphalt Paving Technologists*, 61, 217-240.
- [10] **Krutz, N.C., and Sebaaly, P.E.** (1993) Effect of aggregate gradation on permanent deformation of asphaltic concrete. *Proceedings of Association of Asphalt Paving Technologists*, 62, 450-473.
- [11] **Masad, E., Little, D., Tashman, L., Saadeh, S., Al-Rousan, T., and Sukhwani, R.** (2003). "Evaluation of aggregate characteristics affecting HMA concrete performance."
- [12] **MoRTH**, Specifications for Road and Bridge Works, up gradation of 5thRevision, Ministry Of Road Transport and Highways.

